27 Solve the simultaneous equations

$$x + 3y = 12$$
$$5x - y = 4$$

(1) $x + 3y = 12$ (2) $5x - y = 4$

Multiply all of (2) by 3: $15x - 3y = 12$

1 mark
Eliminate y using opposite coefficients

Add this to (1): $16x = 24$

$$16x = 24$$
$$\div 16 \qquad \div 16$$
$$x = 1.5$$

Substitute in (1):

$$1.5 + 3y = 12$$
$$-1.5 \qquad -1.5$$
$$3y = 10.5$$
$$\div 3 \qquad \div 3$$
$$y = 3.5$$

1 mark
Substitute the value found into an original equation

Check: $1.5 + 3 \times 3.5 = 12$; $5 \times 1.5 - 3.5 = 4$.

$x =$ 1.5

1 mark
Both values correct

$y =$ 3.5

(Total for Question 27 is 3 marks)